

457.568 Statistical Hydrology, Fall 2022

Tue 18:30 PM – 21:20 PM, Building 35 Room 432

INSTRUCTOR Dr. Gi Joo Kim

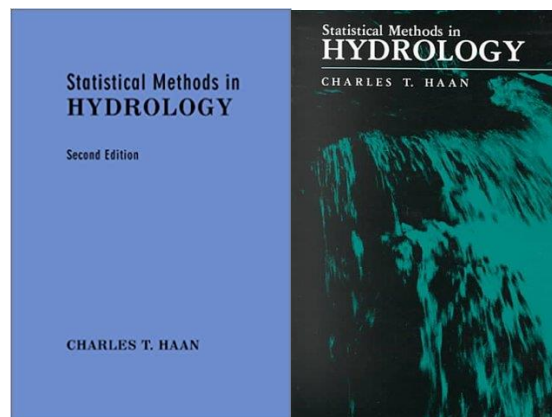
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OFFICE HOURS Email for appointment (generally Tuesday mornings best)

PRIMARY TEXTBOOK

- Hann, C. T. (2002). **Statistical Methods in Hydrology (Second Edition)**. Iowa State Press.



SUGGESTED FURTHER READINGS

- Devore, J. L. (2016). **Probability and Statistics for Engineering and the Sciences, Ninth Edition**. Cengage Learning.
- Chow, V. T. & Maidment, D. R. (1988). **Applied Hydrology**. McGraw-Hill, Inc. (Especially Chapter 11 Hydrologic Statistics & Chapter 12 Frequency Analysis)
- Kottegoda, N. T. & Rosso, R. (1997). **Statistics, Probability, and Reliability for Civil and Environmental Engineers**. McGraw-Hill.
- Teegavarapu, R. S. V., Salas, J. D., & Stedinger, J. R. (2019). **Statistical Analysis of Hydrologic Variables**. ASCE.
- Maidment, D. R. (1993). **Handbook of Hydrology**. McGraw-Hill, Inc. (Especially Chapters 17-20:
 - Chapter 17 Statistical Treatment of Hydrologic Data
 - Chapter 18 Frequency Analysis of Extreme Events
 - Chapter 19 Analysis and Modeling of Hydrologic Time Series
 - Chapter 20 Geostatistics)

COURSE DESCRIPTION

Dealing with uncertain variables especially in the field of hydrology is crucial since all factors included in many of the hydrologic processes are subject to variabilities caused from either random or anthropogenic causes. Generally, the causes are classified into the following three reasons: (1) inherent randomness of the hydrologic system, (2) sampling error, and (3) the result from incorrect understanding of the processes involved. This course enables the students to deal with random hydrological variables with proper methods such as probabilistic modeling, extreme event analysis, regression models, and time series models. Both the assignments and the final project at the end of the semester will challenge the students to deal with real-world hydrological data.

PREREQUISITS

- Basic probability & statistics
 - Coding Ability in R
- ➔ You will be given opportunities to learn coding in R throughout the class, so do not worry so much if you are unfamiliar with R.

TOPICS COVERED

- Review on basic statistics & probability concepts
- Probabilistic modeling methods in hydrology
- Extreme event analysis using various methods (GEV, POT, etc.)
- Trend detection, regression, and time series analysis

GRADING

Participation 10%

Assignments 30% (15% Datacamp Assignments & 15% In-class Assignments)

Quizzes 20%

Test 20%

Mid-Term Project 10%

Final Project 10%

- ⇒ 위의 명시되지 않은 부분에 있어서는 서울대학교 학칙 및 규정에 따름.
- ⇒ 늦게 제출되는 과제/프로젝트의 경우에는
제출기한 이후 3일까지 하루에 -10% (최대 -30%), 4-7일까지 -50%, 7일 초과시 0%.

COURSE SCHEDULE (FALL 2022)

WEEK/DATE	TOPIC	ASSIGNMENTS & QUIZZES
1 Sep-06	<i>Course Introduction</i> Lec 01: Review on Basic Probability Concepts	
2 Sep-13	Lec 02: Probabilistic Modeling in Hydrology- Part I (Moments & Probabilistic Models)	
3 Sep-20	Lec 03: Probabilistic Modeling in Hydrology- Part II (MLE & Goodness of fit tests)	Quiz 1
4 Sep-27	<i>Training with Probabilistic Modeling</i>	
5 Oct-04	Field Trip (Details TBD)	
6 Oct-11	Lec 04: Extreme Event Analysis- Part I (GEV Distribution & Uncertainties)	Assignment 1 Due
7 Oct-18	Lec 05: Extreme Event Analysis- Part II (LP3 Distributions & POT Models)	
8 Oct-25	<i>Term Project Presentation: Bulletin 17B/17C</i>	
9 Nov-01	Lec 06: Extreme Event Analysis- Part III (Selecting thresholds & Estimation using POT models)	Quiz 2
10 Nov-08	<i>Training with Extreme Event Analysis</i> Lec 07: Introduction to Trend Detection	
11 Nov-15	Lec 08: Analysis using Regression Models <i>Training with Regression</i>	
12 Nov-22	TEST (Closed Book)	Assignment 2 Due
13 Nov-29	Lec 09: Introduction to Time Series Analysis	
14 Dec-06	Lec 10: Hydrologic Statistics Frontiers -Introduction and Examples	
15 Dec-13	<i>Final Project Presentation</i>	Final Project Report Due Dec-20